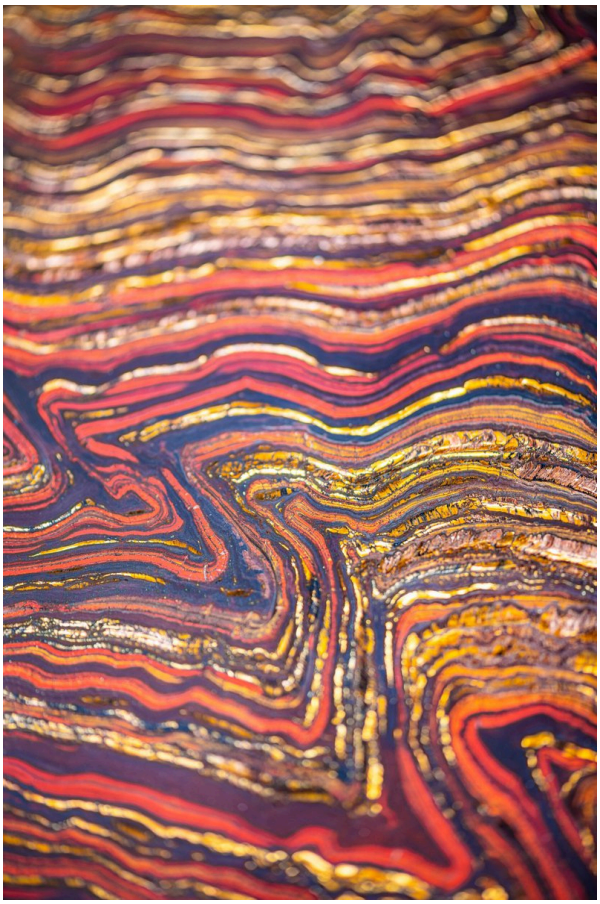
BANDED IRON FORMATION



Banded iron formations are alternating silica and iron rich layer rocks.

Characteristics: Banded iron-formations are sedimentary rock formations with alternating silica-rich layers and iron-rich layers that are typically composed of iron oxides (hematite and magnetite), iron-rich carbonates (siderite and ankerite), and/or iron-rich silicates (e.g., minnesotaite and greenalite).

Natural Occurrences: Banded Iron Formations (BIFs), chemogenic sedimentary rocks, formed from chemical precipitation on ancient seafloors, mostly during the Paleoproterozoic era. Superior-type BIFs on stable continental shelves can exceed 100 meters in thickness and span over 100 kilometers. Composed of hematite and magnetite, these deposits formed as upwelling iron-rich waters reacted with oxygen from early photosynthetic cyanobacteria, with UV radiation possibly aiding iron oxidation. Silica for chert layers, abundant in ancient seawater (~120 mg/l), contrasts with modern levels (<10 mg/l) due to silica-extracting organisms like diatoms and sponges, preventing current BIF formation.

Fun facts: BIFs are economically vital as a primary source of iron, crucial to modern society. Their iron likely originated from oceanic sources like black smokers and dissolved oceanic crust, differing from modern ironstones, which include continental materials.